

Deep Learning for Stock Price Prediction: LSTM vs Transformer with Sentiment Analysis

CS 7643 - Deep Learning

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Abstract

Stock price prediction remains challenging due to market noise and non-stationarity. We investigate whether deep learning models enhanced with sentiment analysis can improve next-day price direction prediction. We implement and train LSTM and Transformer architectures on four tech stocks (AAPL, AMZN, NVDA, TSLA) using technical indicators combined with StockTwits sentiment (2020-2022 training, 2026 validation). Key findings: (1) Both models perform nearly identically (LSTM: 43.57%, Transformer: 44.00%), with Transformer having minimal advantage (+0.43%); (2) severe class imbalance causes prediction collapse, with models predicting predominantly one class; (3) both models perform below random baseline (50%) on 2026 test data; (4) validation-to-test degradation of 13-15 percentage points indicates poor generalization; (5) limited training data (415 samples) insufficient for either architecture. Our analysis demonstrates that financial time series prediction is significantly more challenging than initially apparent, requiring careful attention to class imbalance, data quantity, and evaluation methodology.

1. Introduction

What are we trying to do? Predict whether a stock's price will go up or down tomorrow, using both historical price data and investor sentiment from social media.

Why is this hard? Stock markets are noisy, non-stationary (patterns change over time), and influenced by countless factors. Traditional technical analysis uses only price/volume data, missing important sentiment signals from social media.

Who cares? Accurate price prediction could improve trading strategies and risk assessment. More broadly, understanding how deep learning handles financial time series has applications in economics and algorithmic trading.

Our approach: We train deep learning models (LSTM and Transformer) to learn temporal patterns from sequences of financial and sentiment features. We compare architectures, analyze failure modes, and validate generalization on 2026 out-of-sample data.

Key contributions:

- Rigorous comparison of LSTM vs Transformer on financial time series

- Analysis of class imbalance and prediction collapse in stock prediction
- Demonstration of poor generalization across market regimes
- Evidence that limited data (415 samples) insufficient for deep learning
- Honest reporting of below-baseline results with methodological insights

2. Related Work

Recurrent architectures (LSTM, GRU) are popular for financial prediction due to their ability to capture temporal dependencies. **Attention mechanisms** and Transformers have shown promise in NLP and are increasingly applied to time series. **Sentiment analysis** from social media (Twitter, StockTwits) has been shown to provide predictive signals for stock movements. Our work extends prior research by: (1) implementing Transformer from scratch for financial prediction, (2) rigorous train/validation/test methodology, (3) comprehensive analysis of failure modes including class imbalance and prediction collapse, and (4) honest reporting of negative results.

3. Data

3.1 Dataset Description

We collected daily data for four large-cap technology stocks: AAPL, AMZN, NVDA, and TSLA.

Training: 2020-2022 (755 trading days, 80/20 train/val split)

Validation: 2020-2022 (20% holdout, 104 samples per ticker)

Test: 2026 (53-58 trading days, out-of-sample)

Total: 2,600 stock-days across 4 tickers

3.2 Data Collection

Market Data: Yahoo Finance API provides adjusted closing prices, volume, and OHLC data. We compute 17 technical indicators including RSI, MACD, Bollinger Bands, and rolling returns.

StockTwits Sentiment: Public API provides social media messages with user-labeled sentiment (bullish/bearish). We score messages using RoBERTa (cardiffnlp/twitter-roberta-base-sentiment-latest) and aggregate daily: message count, sentiment mean/std, and sum statistics. Coverage: 755 days (100%).

3.3 Features and Target

Input Features (31 total):

- Technical (17): Price, returns, volatility, RSI, MACD, Bollinger Bands, volume
- Sentiment (14): StockTwits metrics (count, mean, std, sum)

Target: Binary classification - UP (1) if $P_{t+1} > P_t$, DOWN (0) otherwise.

Sequence Construction: Rolling windows of 5 days (31 features $\times$ 5 timesteps) predict next-day direction.

3.4 Data Limitations

- Only 4 tech stocks (may not generalize to other sectors)
- Limited training data (415 samples after 80/20 split)
- Survivorship bias (only large-cap stocks)
- Market regime: trained on bull market (2020-2022), tested on 2026

4. Methodology

4.1 LSTM Architecture

Architecture:

- Input: 31 features $\times$ 5 timesteps
- LSTM: 2 layers, 128 hidden units, 20% dropout
- Dense: 64 units, ReLU activation, 20% dropout
- Output: 1 unit, Sigmoid activation

Parameters: 200K trainable

4.2 Transformer Architecture

Architecture:

- Input projection: 31 $\rightarrow$ 64 dimensions
- Positional encoding: Sinusoidal
- Transformer encoder: 2 layers, 4 attention heads
- Feed-forward: 128 dimensions per layer
- Global average pooling
- Classification head: 64 $\rightarrow$ 32 $\rightarrow$ 1

Parameters: 150K trainable

4.3 Training Details

Train/Val/Test Split:

- Split 2020-2022 into train (80%, 415 samples) and validation (20%, 104 samples)

- Select best model based on validation accuracy

- Evaluate on 2026 test data only once at the end

Loss Function: Binary cross-entropy

Optimizer: Adam with learning rate 0.001

Regularization:

- Dropout: 20% in LSTM, 10% in Transformer
- Early stopping: patience=10 epochs on validation loss

Hyperparameters:

- Sequence length: 5 days
- Batch size: 32
- Max epochs: 50 (early stopping typically at 10-20)

5. Results

5.1 Overall Performance

Model	Validation	Test (2026)
LSTM	58.63%	43.57%
Transformer	57.43%	44.00%
Random Baseline	50.00%	50.00%

Table 1: Overall model performance. Both models perform below random baseline on 2026 test data.

Key Finding: Both models perform nearly identically, with Transformer having minimal advantage (+0.43%). More critically, both perform below 50% random baseline on test data.

5.2 Per-Ticker Results

Ticker	LSTM		Transformer	
	Val	Test	Val	Test
AAPL	59.62%	48.28%	58.65%	48.28%
AMZN	53.33%	41.38%	52.38%	41.38%
NVDA	62.50%	41.51%	62.50%	41.51%
TSLA	59.05%	43.10%	56.19%	44.83%
Avg	58.63%	43.57%	57.43%	44.00%

Table 2: Per-ticker performance. Models perform identically on AAPL, AMZN, and NVDA. Only TSLA shows difference (+1.73% for Transformer).

Analysis:

- AAPL, AMZN, NVDA: Identical performance (both models predict same class)
- TSLA: Transformer slightly better (+1.73%)
- All tickers: Massive validation-to-test degradation (-11% to -21%)

5.3 Class Imbalance Analysis

Both models suffer from severe class imbalance:

AAPL, AMZN, NVDA (both models):

- Down: 100% recall, 41-48% precision
- Up: 0% recall, 0% precision
- **Models always predict "Down"**

TSLA:

- LSTM: Down 88% recall, Up 6% recall
- Transformer: Down 100% recall, Up 0% recall
- **Models predominantly predict "Down"**

Why this happens:

- Test period has class imbalance (more "Down" days)
- Models learn to predict majority class
- Achieves reasonable accuracy but no predictive value

5.4 Generalization Analysis

Ticker	LSTM Deg.	Trans. Deg.
AAPL	-11.34 pp	-10.37 pp
AMZN	-11.95 pp	-11.00 pp
NVDA	-20.99 pp	-20.99 pp
TSLA	-15.95 pp	-11.36 pp
Average	-15.06 pp	-13.43 pp

Table 3: Validation-to-test degradation. Both models show massive performance drop, indicating poor generalization to 2026 market conditions.

Analysis: 13-15 percentage point degradation indicates:

- Market regime change between 2020-2022 and 2026
- Models overfitting to training period patterns
- Insufficient regularization or data diversity

6. Discussion

6.1 Why Both Models Fail

1. Insufficient Training Data

- Only 415 training samples after 80/20 split
- Deep learning typically requires 10K-100K samples
- Both LSTM and Transformer underperform

2. Severe Class Imbalance

- Models exploit test set class distribution
- Predict majority class to minimize loss
- No real pattern learning occurs

3. Market Regime Change

- Training: 2020-2022 bull market
- Testing: 2026 mixed conditions
- Patterns don't transfer across regimes

4. Limited Signal in Sentiment

- Sentiment alone insufficient for prediction
- Need additional features (news, fundamentals, macro)
- Social media sentiment may be noisy

6.2 Why Models Perform Identically

For 3 out of 4 tickers, both models achieve identical accuracy because:

1. Both predict only the "Down" class
2. Accuracy equals proportion of "Down" samples in test set
3. No architectural advantage when prediction collapses

Only TSLA shows difference because Transformer maintains slightly more balanced predictions.

6.3 Comparison to Literature

Prior work reports 52-58% accuracy on stock prediction tasks. Our results (43-44%) are below this range, likely due to:

- Stricter evaluation (proper train/val/test split)
- Limited training data
- Challenging test period (2026 market conditions)
- No data leakage or look-ahead bias

6.4 Limitations

1. **Limited scope:** Only 4 tech stocks
2. **Insufficient data:** 415 training samples too few
3. **Short horizon:** Next-day prediction only
4. **Class imbalance:** Not addressed in training
5. **No transaction costs:** Real trading has fees
6. **Single sentiment source:** Only StockTwits

7. Lessons Learned

7.1 Methodological Insights

1. Proper Train/Val/Test Split is Critical

- Validation set must be used for model selection
- Test set should be "locked box" until final evaluation
- Prevents overfitting to test set patterns

2. Class Imbalance Must Be Addressed

- Use class weights in loss function
- Try focal loss or resampling techniques
- Monitor per-class metrics, not just accuracy

3. Multiple Metrics Required

- Accuracy alone insufficient
- Check precision, recall, F1 per class
- Monitor prediction distribution

7.2 Practical Recommendations

For Future Work:

1. **Collect more data:** Need 10-100x more samples
2. **Address class imbalance:** Use class weights or focal loss
3. **Try simpler baselines:** Logistic regression, random forest
4. **Add more features:** News, fundamentals, market-wide indicators
5. **Ensemble methods:** Combine multiple models
6. **Market regime detection:** Separate models for different conditions

7.3 When to Use LSTM vs Transformer

Based on our results:

Use LSTM when:

- Limited training data (< 10K samples)
- Strong temporal dependencies
- Need stable training

Use Transformer when:

- Large training data (> 100K samples)
- Need interpretability (attention weights)
- Parallel processing beneficial

Our case: With 415 samples, neither architecture has sufficient data. Performance is nearly identical.

8. Conclusion

We implemented and trained LSTM and Transformer architectures for stock price prediction with sentiment analysis. Our findings:

Key Results:

1. **Both models perform identically** (LSTM: 43.57%, Transformer: 44.00%)
2. **Below random baseline** (50%) on 2026 test data
3. **Severe class imbalance** causes prediction collapse
4. **Poor generalization** (-13% to -15% degradation)
5. **Insufficient training data** (415 samples too few)

Insights:

- Financial time series prediction is harder than initially apparent
- Deep learning requires significantly more data than available
- Class imbalance must be addressed before meaningful progress
- Proper validation methodology is critical
- Negative results are valuable for the field

Future Work:

- Collect 10-100x more training data
- Address class imbalance with focal loss or resampling
- Try simpler baselines (logistic regression, random forest)
- Add more features (news, fundamentals, macro indicators)
- Explore ensemble methods and transfer learning

Our work demonstrates that rigorous evaluation methodology and honest reporting of negative results are essential for advancing deep learning research. The fact that both sophisticated architectures perform below baseline highlights the fundamental challenges in financial time series prediction and the importance of data quantity and quality.

Acknowledgments

We thank the CS 7643 teaching staff for guidance throughout this project. StockTwits data collected via public API. RoBERTa sentiment model from HuggingFace.

Code Repository

All code, trained models, and data available at:
<https://github.com/KKONAM/financial-sentiment-analysis-spring-2026/tree/transformer-model-2026>

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Team Contributions

Team Member	Contributions
Agha A Raza	• Implemented Transformer architecture from scratch • Designed and conducted experiments • Analyzed class imbalance and prediction collapse • Wrote methodology and analysis sections • Created training visualizations
Konam G Kenneth	• Implemented LSTM baseline architecture • Collected and processed StockTwits data with RoBERTa • Designed proper train/validation/test methodology • Conducted 2026 out-of-sample validation • Wrote data collection and results sections
Shared	• Hyperparameter tuning and model training • Analysis of results and failure modes • Paper writing and editing • Code repository organization • Figure generation and visualization

Note: Both team members contributed equally to the technical implementation, experimentation, and analysis.